MTE 204 Problem Set #3: Solutions

Systems of Linear Equations (Chapter 11 in Chapra 8th edition, International Version)

11.9 Recall from Prob. 10.8, that the following system of equations is designed to determine concentrations (the c's in g/m^3) in a series of coupled reactors as a function of amount of mass input to
each reactor (the right-hand sides are in g/day):

$$15c_1 - 3c_2 - c_3 = 3300$$
$$-3c_1 + 18c_2 - 6c_3 = 1200$$
$$-4c_1 - c_2 + 12c_3 = 2400$$

Solve this problem with the Gauss-Seidel method to $\varepsilon_s = 5\%$.

The first iteration can be implemented as

$$c_1 = \frac{3300 + 3c_2 + c_3}{15} = \frac{3300 + 3(0) + 0}{15} = 220.00$$

$$c_2 = \frac{1200 + 3c_1 + 6c_3}{18} = \frac{1200 + 3(220.00) + 6(0)}{18} = 103.33$$

$$c_3 = \frac{2400 + 4c_1 + c_2}{12} = \frac{2400 + 4(220.00) + 103.33}{12} = 281.94$$

Second iteration:

$$c_1 = \frac{3300 + 3c_2 + c_3}{15} = \frac{3300 + 3(103.33) + 281.94}{15} = 259.46$$

$$c_2 = \frac{1200 + 3c_1 + 6c_3}{18} = \frac{1200 + 3(259.46) + 6(281.94)}{18} = 203.89$$

$$c_3 = \frac{2400 + 4c_1 + c_2}{12} = \frac{2350 + 4(259.46) + 203.89}{12} = 303.48$$

The error estimates for the second iteration can be computed as

$$\varepsilon_{a,1} = \left| \frac{259.46 - 220.00}{259.46} \right| \times 100\% = 15.21\%$$

$$\varepsilon_{a,2} = \left|\frac{203.89 - 103.33}{203.89}\right| \times 100\% = 49.32\%$$

$$\varepsilon_{a,3} = \left|\frac{303.48 - 281.94}{303.48}\right| \times 100\% = 7.10\%$$

The remainder of the calculation proceeds until all the errors fall below the stopping criterion of 5%. The entire computation can be summarized as

i	c_1	$\varepsilon_{a,1}$	c_2	$\varepsilon_{a,2}$	c_3	$\varepsilon_{a,3}$	max ε_a
0	0		0		0		
1	220.00	100.00	103.33	100.00	281.94	100.00	100.00
2	259.46	15.21	203.89	49.32	303.48	7.10	49.32
3	281.01	7.67	214.66	5.02	311.56	2.59	7.67
4	283.70	0.95	217.80	1.44	312.72	0.37	1.44

Thus, after 4 iterations, the maximum error is 1.44% and we arrive at the result: c_1 = 283.70, c_2 = 217.80 and c_3 = 312.72. Note that after 10 iterations, the process would converge on c_1 = 284.56, c_2 = 218.45 and c_3 = 313.06.

11.11 Use the Gauss-Seidel method to solve the following system until the percent relative error falls below ε_s = 5%:

$$\begin{aligned} 10x_1 + 2x_2 - x_3 &= 22 \\ -3x_1 - 6x_2 + 2x_3 &= -14 \\ x_1 + x_2 + 5x_3 &= 14 \end{aligned}$$

The first iteration can be implemented as

$$x_1 = \frac{22-2x_2+x_3}{10} = \frac{22-2(0)+0}{10} = 2.2$$
$$x_2 = \frac{-14+3x_1-2x_3}{-6} = \frac{-14+3(2.2)-2(0)}{-6} = 1.23$$

$$x_3 = \frac{14-x_1-x_2}{5} = \frac{14-(2.2)-1.23}{5} = 2.11$$

Second iteration:

$$x_1 = \frac{22-2(1.23)-2.11}{10} = 2.16$$
$$x_2 = \frac{-14+3(2.16)-2(2.11)}{-6} = 1.96$$

$$x_3 = \frac{14-(2.16)-1.96}{5} = 1.98$$

The error estimates can be computed as

$$\varepsilon_{a,1} = \left|\frac{2.16-2.20}{2.16}\right| \times 100\% = 1.63\%$$
$$\varepsilon_{a,2} = \left|\frac{1.96-1.23}{1.96}\right| \times 100\% = 36.93\%$$

$$\varepsilon_{a,3} = \left|\frac{1.98-(2.11)}{1.98}\right| \times 100\% = 6.95\%$$

The remainder of the calculation proceeds until all the errors fall below the stopping criterion of 5%. The entire computation can be summarized as

i	x_1	$\varepsilon_{a,1}$	x_2	$\varepsilon_{a,2}$	x_3	$\varepsilon_{a,3}$	max ε_a
0	0		0		0		
1	2.20	100.00	1.23	100.00	2.11	100.00	100.00
2	2.16	1.63	1.96	36.93	1.98	6.95	36.93
3	2.01	7.88	1.99	1.67	2.00	1.25	7.88
4	2.00	0.21	2.00	0.52	2.00	0.06	0.52

Thus, after 5 iterations, the maximum error is 0.59% and we arrive at the result: $x_1 = 2.00$, $x_2 = 2.00$ and $x_3 = 2.00$. Note that the exact result is [2 2 2].

11.12 Use the Gauss-Seidel method **(a)** without relaxation and **(b)** with relaxation (λ = 0.95) to solve the following system to a tolerance of ε_s = 5%. If necessary, rearrange the equations to achieve convergence.

$$-3x_1 + x_2 + 15x_3 = 44$$
$$6x_1 - 2x_2 + x_3 = 5$$
$$5x_1 + 10x_2 + x_3 = 28$$

The equations should first be rearranged so that they are diagonally dominant,

$$6x_1 - 2x_2 + x_3 = 5$$
$$5x_1 + 10x_2 + x_3 = 28$$
$$-3x_1 + x_2 + 15x_3 = 44$$

Each can be solved for the unknown on the diagonal as

$$x_1 = \frac{5 + 2x_2 - x_3}{6}$$
$$x_2 = \frac{28 - 5x_1 - x_3}{10}$$
$$x_3 = \frac{44 + 3x_1 - x_2}{15}$$

(a) The first iteration can be implemented as

$$x_1 = \frac{5 + 2(0) - 0}{6} = 0.83333$$
$$x_2 = \frac{28 - 5(0.83333) - 0}{10} = 2.38333$$
$$x_3 = \frac{44 + 3(0.83333) - 2.38333}{15} = 2.94111$$

Second iteration:

$$x_1 = \frac{5 + 2(2.38333) - 2.94111}{6} = 1.13759$$
$$x_2 = \frac{28 - 5(1.13759) - 2.94111}{10} = 1.93709$$
$$x_3 = \frac{44 + 3(1.13759) - 1.93709}{15} = 3.03171$$

The error estimates can be computed as

$$\varepsilon_{a,1} = \left| \frac{1.13759 - 0.83333}{1.13759} \right| \times 100\% = 26.75\%$$

$$\varepsilon_{a,2} = \left| \frac{1.93709 - 2.38333}{1.93709} \right| \times 100\% = 23.04\%$$

$$\varepsilon_{a,3} = \left| \frac{3.03171 - 2.94111}{3.03171} \right| \times 100\% = 2.99\%$$

The remainder of the calculation proceeds until all the errors fall below the stopping criterion of 5%. The entire computation can be summarized as

i	x_1	$\varepsilon_{a,1}$	x_2	$\varepsilon_{a,2}$	x_3	$\varepsilon_{a,3}$	max ε_a
0	0		0		0		
1	0.83333	100.00	2.38333	100.00	2.94111	100.00	100.00
2	1.13759	26.75	1.93709	23.04	3.03171	2.99	26.75
3	0.97375	16.83	2.00996	3.63	2.99409	1.26	16.83
4	1.00430	3.04	1.99844	0.58	3.00096	0.23	3.04

Thus, after 4 iterations, the maximum error is 3.04% and we arrive at the result: x_1 = 1.00430, x_2 = 1.99844 and x_3 = 3.00096. The exact solution, achieved in 22 iterations where the maximum error is below machine precision, but practically achieved in 10 iterations, is [1 2 3].

(b) Our recursion equations for relaxation in Gauss-Seidel are

$$x_1 = \frac{5 + 2x_{2r}^{old} - x_{3r}^{old}}{6}$$
$$x_{1r} = \lambda x_1 + (1-\lambda) x_{1r}^{old}$$
$$x_2 = \frac{28 - 5x_{1r} - x_{3r}^{old}}{10}$$
$$x_{2r} = \lambda x_2 + (1-\lambda) x_{2r}^{old}$$
$$x_3 = \frac{44 + 3x_{1r} - x_{2r}}{15}$$
$$x_{3r} = \lambda x_3 + (1-\lambda) x_{3r}^{old}$$

<u>First iteration:</u> To start, assume $x_1 = x_{1r}^{old} = x_2 = x_{2r}^{old} = x_3 = x_{3r}^{old} = 0$

$$x_1 = \frac{5 + 2(0) - 0}{6} = 0.83333$$

Apply relaxation:

$$x_{1r} = 0.95(0.83333) + (1-0.95)0 = 0.791667$$

$$x_2 = \frac{28 - 5(0.791667) - 0}{10} = 2.404167$$

$$x_{2r} = 0.95(2.404167) + (1-0.95)0 = 2.283958$$

$$x_3 = \frac{44 + 3(0.791667) - 2.283958}{15} = 2.939403$$

$$x_{3r} = 0.95(2.939403) + (1-0.95)0 = 2.792433$$

Note that error estimates are not necessary on the first iteration, as all errors will be 100%; however, in developing a spreadsheet or macro, these are useful values to calculate as a check. If something other than 100% is found, it indicates an error in programming.

<u>Second iteration:</u>

$$x_1 = \frac{5 + 2(2.283958) - 2.792433}{6} = 1.129247$$

$$x_1 = 0.95(1.129247) + (1 - 0.95)(0.791667) = 1.112368$$

At this point, an error estimate can be made

$$\varepsilon_{a,1} = \left| \frac{1.112368 - 0.791667}{1.112368} \right| 100\% = 28.83\%$$

Because this error exceeds the stopping criterion, it will not be necessary to compute error estimates for the remainder of this iteration.

$$x_2 = \frac{28 - 5(0.791667) - 2.792433}{10} = 1.964573$$

$$x_{2r} = 0.95(1.964573) + (1 - 0.95)2.283958 = 1.980542$$

$$x_3 = \frac{44 + 3(1.112368) - 1.980542}{15} = 3.023771$$

$$x_{3r} = 0.95(3.023771) + (1 - 0.95)2.792433 = 3.012204$$

The computations can be continued for one more iteration. The entire calculation is summarized in the following table.

i	x_1	x_{1r}	ε_{a1} (%)	x_2	x_{2r}	ε_{a2} (%)	x_3	x_{3r}	ε_{a3} (%)
0	0	0		0	0		0	0	
1	0.833333	0.791667	100.00	2.404167	2.283958	100.00	2.939403	2.792433	100.00
2	1.129247	1.112368	28.83	1.964573	1.980542	15.32	3.023771	3.012204	7.30
3	0.99148	0.997524	11.51	2.000017	1.999044	0.93	2.999569	3.0002	0.40
4	0.999648	0.999542	0.20	2.000209	2.000151	0.06	2.999898	2.999913	0.01

After 3 iterations, the approximate errors fall below the stopping criterion with the final result: x_{1r} = 0.999542, x_{2r} = 2.000151 and x_{3r} = 2.999913. Note that the relaxed values converge on the exact solution after 15 iterations at $x_1 = 1$, $x_2 = 2$ and $x_3 = 3$. This is significantly faster than the 22 iterations of the non-relaxed method.

11.13 Use the Gauss-Seidel method **(a)** without relaxation and **(b)** with relaxation ($\lambda = 1.2$) to solve the following system to a tolerance of $\varepsilon_s = 5\%$. If necessary, rearrange the equations to achieve convergence.

$$
\begin{aligned}
2x_1 - 6x_2 - x_3 &= -38 \\
-3x_1 - x_2 + 7x_3 &= -34 \\
-8x_1 + x_2 - 2x_3 &= -20
\end{aligned}
$$

The equations must first be rearranged so that they are diagonally dominant

$$
\begin{aligned}
-8x_1 + x_2 - 2x_3 &= -20 \\
2x_1 - 6x_2 - x_3 &= -38 \\
-3x_1 - x_2 + 7x_3 &= -34
\end{aligned}
$$

(a) The first iteration can be implemented as

$$x_1 = \frac{-20 - x_2 + 2x_3}{-8} = \frac{-20 - 0 + 2(0)}{-8} = 2.5$$

$$x_2 = \frac{-38 - 2x_1 + x_3}{-6} = \frac{-38 - 2(2.5) + 0}{-6} = 7.166667$$

$$x_3 = \frac{-34 + 3x_1 + x_2}{7} = \frac{-34 + 3(2.5) + 7.166667}{7} = -2.761905$$

Second iteration:

$$x_1 = \frac{-20 - 7.166667 + 2(-2.761905)}{-8} = 4.08631$$

$$x_2 = \frac{-38 - 2x_1 + x_3}{-6} = \frac{-38 - 2(4.08631) + (-2.761905)}{-6} = 8.155754$$

$$x_3 = \frac{-34 + 3x_1 + x_2}{7} = \frac{-34 + 3(4.08631) + 8.155754}{7} = -1.94076$$

The error estimates can be computed as

$$\varepsilon_{a,1} = \left|\frac{4.08631 - 2.5}{4.08631}\right| \times 100\% = 38.82\%$$

$$\varepsilon_{a,2} = \left|\frac{8.155754 - 7.166667}{8.155754}\right| \times 100\% = 12.13\%$$

$$\varepsilon_{a,3} = \left|\frac{-1.94076 - (-2.761905)}{-1.94076}\right| \times 100\% = 42.31\%$$

The remainder of the calculation proceeds until all the errors fall below the stopping criterion of 5%. The entire computation can be summarized as

iteration	unknown	value	ε_a	maximum ε_a
0	$x1$	0		
	$x2$	0		
	$x3$	0		
1	$x1$	2.5	100.00%	
	$x2$	7.166667	100.00%	
	$x3$	-2.7619	100.00%	100.00%
2	$x1$	4.08631	38.82%	
	$x2$	8.155754	12.13%	
	$x3$	-1.94076	42.31%	42.31%
3	$x1$	4.004659	2.04%	
	$x2$	7.99168	2.05%	
	$x3$	-1.99919	2.92%	2.92%

Thus, after 3 iterations, the maximum error is 2.92% and we arrive at the result: $x_1 = 4.004659$, $x_2 = 7.99168$ and $x_3 = -1.99919$.

(b) The same computation can be developed with relaxation where $\lambda = 1.2$.

First iteration:

$$x_1 = \frac{-20 - x_2 + 2x_3}{-8} = \frac{-20 - 0 + 2(0)}{-8} = 2.5$$

Relaxation yields: $x_1 = 1.2(2.5) - 0.2(0) = 3$

$$x_2 = \frac{-38 - 2x_1 + x_3}{-6} = \frac{-38 - 2(3) + 0}{-6} = 7.333333$$

Relaxation yields: $x_2 = 1.2(7.333333) - 0.2(0) = 8.8$

$$x_3 = \frac{-34 + 3x_1 + x_2}{7} = \frac{-34 + 3(3) + 8.8}{7} = -2.3142857$$

Relaxation yields: $x_3 = 1.2(-2.3142857) - 0.2(0) = -2.7771429$

Second iteration:

$$x_1 = \frac{-20 - x_2 + 2x_3}{-8} = \frac{-20 - 8.8 + 2(-2.7771429)}{-8} = 4.2942857$$

Relaxation yields: $x_1 = 1.2(4.2942857) - 0.2(3) = 4.5531429$

$$x_2 = \frac{-38 - 2x_1 + x_3}{-6} = \frac{-38 - 2(4.5531429) - 2.7771429}{-6} = 8.3139048$$

Relaxation yields: $x_2 = 1.2(8.3139048) - 0.2(8.8) = 8.2166857$

$$x_3 = \frac{-34 + 3x_1 + x_2}{7} = \frac{-34 + 3(4.5531429) + 8.2166857}{7} = -1.7319837$$

Relaxation yields: $x_3 = 1.2(-1.7319837) - 0.2(-2.7771429) = -1.5229518$

The error estimates can be computed as

$$\varepsilon_{a,1} = \left|\frac{4.5531429 - 3}{4.5531429}\right| \times 100\% = 34.11\%$$

$$\varepsilon_{a,2} = \left|\frac{8.2166857 - 8.8}{8.2166857}\right| \times 100\% = 7.1\%$$

$$\varepsilon_{a,3} = \left|\frac{-1.5229518 - (-2.7771429)}{-1.5229518}\right| \times 100\% = 82.35\%$$

The remainder of the calculation proceeds until all the errors fall below the stopping criterion of 5%. The entire computation can be summarized as

iteration	unknown	value	relaxation	ε_a	maximum ε_a

1	$x1$	2.5	3	100.00%	
	$x2$	7.3333333	8.8	100.00%	
	$x3$	-2.314286	-2.777143	100.00%	100.000%
2	$x1$	4.2942857	4.5531429	34.11%	
	$x2$	8.3139048	8.2166857	7.10%	
	$x3$	-1.731984	-1.522952	82.35%	82.353%
3	$x1$	3.9078237	3.7787598	20.49%	
	$x2$	7.8467453	7.7727572	5.71%	
	$x3$	-2.12728	-2.248146	32.26%	32.257%
4	$x1$	4.0336312	4.0846055	7.49%	
	$x2$	8.0695595	8.12892	4.38%	
	$x3$	-1.945323	-1.884759	19.28%	19.280%
5	$x1$	3.9873047	3.9678445	2.94%	
	$x2$	7.9700747	7.9383056	2.40%	
	$x3$	-2.022594	-2.050162	8.07%	8.068%
6	$x1$	4.0048286	4.0122254	1.11%	
	$x2$	8.0124354	8.0272613	1.11%	
	$x3$	-1.990866	-1.979007	3.60%	3.595%

Thus, relaxation actually seems to retard convergence. After 6 iterations, the maximum error is 3.595% and we arrive at the result: $x_1 = 4.0122254$, $x_2 = 8.0272613$ and $x_3 = -1.979007$.

11.18 An electronics company produces transistors, resistors, and computer chips. Each transistor requires four units of copper, one unit of zinc, and two units of glass. Each resistor requires three, three, and one units of the three materials, respectively, and each computer chip requires two, one, and three units of these materials, respectively. Putting this information into table form, we get:

Component	Copper	Zinc	Glass
Transistors	4	1	2
Resistors	3	3	1
Computer chips	2	1	3

Supplies of these materials vary from week to week, so the company needs to determine a different production run each week. For example, one week the total amounts of materials available are 960 units of copper, 510 units of zinc, and 610 units of glass. Set up the system of equations modeling the production run, and use Excel, MATLAB, or Mathcad to solve for the number of transistors, resistors, and computer chips to be manufactured this week.

Define the quantity of transistors, resistors, and computer chips as x_1, x_2 and x_3. The system equations can then be defined as

$$
\begin{aligned}
4x_1 + 3x_2 + 2x_3 &= 960 \\
x_1 + 3x_2 + x_3 &= 510 \\
2x_1 + x_2 + 3x_3 &= 610
\end{aligned}
$$

The solution can be implemented in Excel as shown below:

	A	B	C	D
1	A:			B:
2	4	3	2	960
3	1	3	1	510
4	2	1	3	610
5				
6	AI:			X:
7	0.421053	-0.36842	-0.15789	120
8	-0.05263	0.421053	-0.10526	100
9	-0.26316	0.105263	0.473684	90

The following view shows the formulas that are employed to determine the inverse in cells A7:C9 and the solution in cells D7:D9.

	A	B	C	D
1	A:			B:
2	4	3	2	960
3	1	3	1	510
4	2	1	3	610
5				
6	AI:			X:
7	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MMULT(A7:C9,D2:D4)
8	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MMULT(A7:C9,D2:D4)
9	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MINVERSE(A2:C4)	=MMULT(A7:C9,D2:D4)

Here is the same solution generated in MATLAB:

```
>> A=[4 3 2;1 3 1;2 1 3];
>> B=[960 510 610]';
>> x=A\B
```

```
x =
   120
   100
    90
```

In both cases, the answer is $x_1 = 120$, $x_2 = 100$, and $x_3 = 90$